

The Imaginary Octonions Are the Unique Non-Lie Malcev Algebra in the Hurwitz Chain

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Theorem. Among the commutator algebras A^- of the real normed division algebras $A \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}\}$ — the complete list, by Hurwitz's theorem — and among their Cayley-Dickson continuations beyond $\mathbb{O}$, exactly one algebra is a genuine (non-Lie) Malcev algebra: $\text{Im}(\mathbb{O})^-$, the imaginary octonions under commutation.

1. Definitions

A **Malcev algebra** is an anticommutative algebra $(M, [\cdot, \cdot])$ satisfying

$$J(x, y, [x, z]) = [J(x, y, z), x], \quad J(x, y, z) := [[x, y], z] + [[y, z], x] + [[z, x], y].$$

Every Lie algebra ($J \equiv 0$) trivially satisfies this — the Malcev identity is a strict weakening of Jacobi, not an alternative to it. We call M **non-Lie Malcev** if it satisfies the Malcev identity *and* $J \not\equiv 0$. This distinction is essential: Malcev-satisfaction alone does not certify anything structurally interesting, since it is satisfied vacuously by every associative algebra's commutator.

Sagle's theorem (1961): if A is an alternative algebra, then A^- (with $[x, y] := xy - yx$) is a Malcev algebra.

Hurwitz's theorem (1898): the only normed division algebras over $\mathbb{R}$ are $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$, of dimensions 1, 2, 4, 8 — a closed, complete classification, not an open-ended family.

2. The classification, verified

All results below were obtained by direct, independent computation on an octonion/sedenion algebra constructed from first principles via genuine Cayley-Dickson doubling, rather than assumed from prior citation.

$\mathbb{R}, \mathbb{C}$. Both are commutative, so $[x, y] = 0$ identically; the commutator algebra is the trivial (zero) algebra. Vacuously Lie, not non-Lie Malcev.

$\mathbb{H}$. Associative. For any associative algebra, $J(x, y, z) \equiv 0$ identically — this is not merely typically true but a direct algebraic consequence of associativity, verified here by direct computation across randomized triples with zero deviation. $\text{Im}(\mathbb{H})^-$ is the familiar 3-dimensional Lie algebra $\mathfrak{su}(2)$: genuine and non-trivial as a Lie algebra, but Lie, not non-Lie Malcev.

$\mathbb{O}$. Non-associative, but alternative (Hurwitz's theorem forces this: the norm form on a composition algebra of this type enforces the alternative laws). By Sagle's theorem, $\text{Im}(\mathbb{O})^-$ is Malcev. Direct computation confirms both that the Malcev identity holds exactly (tested against 100 random rational combinations of generators, zero violations)

and that the Jacobiator is genuinely non-zero. $\text{Im}(\mathbb{O})^-$ is the unique 7-dimensional simple non-Lie Malcev algebra.

****Beyond $\mathbb{O}$ (sedenions $\mathbb{S}$ and further Cayley-Dickson doublings).**** Hurwitz's theorem terminates the division-algebra chain at $\mathbb{O}$; $\mathbb{S}$ has zero divisors and, critically, is not alternative — so Sagle's theorem no longer applies, and no guarantee of Malcev-preservation survives. Direct computation confirms the stronger fact: $\text{Im}(\mathbb{S})^-$ genuinely *fails* the Malcev identity outright (tested against 100 random combinations, violations found). This is not merely "unproven for $\mathbb{S}$ " — it is proven false.

3. The theorem, and why it is exactly this narrow

Algebra	Associative?	Alternative?	$J \equiv 0$?	Malcev?	Non-Lie Malcev?
$\mathbb{R}^-, \mathbb{C}^-$	—	—	trivially	trivially	No
$\mathbb{H}^- (\cong \mathfrak{su}(2))$	Yes	Yes	Yes	Yes	No
$\mathbb{O}^- (= \text{Im}(\mathbb{O})^-)$	No	Yes	No	Yes	Yes
$\mathbb{S}^-$ and beyond	No	No	No	No	No

The result is not an accident of where the classification happens to stop, but a direct consequence of a single structural fact: associativity forces $J \equiv 0$ (ruling out $\mathbb{R}, \mathbb{C}, \mathbb{H}$), while loss of alternativity removes the one theorem (Sagle's) that could guarantee Malcev survives non-associativity at all (ruling out $\mathbb{S}$ and beyond, confirmed here not to survive it). Octonions occupy the unique position in the chain satisfying both requirements simultaneously: non-associative enough for $J \not\equiv 0$, alternative enough for Sagle's theorem to apply. Given Hurwitz's theorem closes the division-algebra chain at exactly this point, this is not a search that could have turned up a second example further along; the classification is complete by construction.

4. Remark

This specific algebra, $\text{Im}(\mathbb{O})^-$, is the one employed by Günaydin, Lüst & Malek (arXiv:1607.06474) as the origin of the algebraic R-flux structure of non-geometric string theory, and analyzed further in several companion notes to this one.

References

- Günaydin, M., Lüst, D. & Malek, E., "Non-associativity in non-geometric string and M-theory backgrounds, the algebra of octonions, and missing momentum modes," arXiv:1607.06474.
- Sagle, A. A., "Malcev algebras," *Trans. Amer. Math. Soc.* 101 (1961), 426–458.